

Impulse and Impulsive force

§ Impulse of a force:-

The change of momentum during any time is equal to the impulse of the external force and takes place in the direction of the force.

$$I = m(v - u), \quad m = \text{mass of the particle}$$

$v = \text{velocity at time } t$
 $u = \text{initial velocity.}$

Also $I = P \times t$

$P = \text{force}$
 $t = \text{time}$

§ Impulsive force:-

An impulsive force is a very large force which acting on a body for an infinitely small time produces in it a finite change of momentum but the displacement of the body during this time is negligibly small.

§ Principle of conservation of linear momentum:-

The algebraic sum of the momenta of a system of mutually impinging bodies in any given direction is unaltered by their mutual action.

§ Impulse and Work done:-

Let a force P act on a particle of mass m for a time t , during which the particle moves through a distance x and its velocity changes from u to v .

The work done by the force

$$= Px$$
$$= mfx \quad [P = mf]$$

$$= m \cdot \frac{1}{2} (v^2 - u^2) \quad \text{Since } v^2 = u^2 + 2fx$$

$$= m \cdot \frac{1}{2} (v + u)(v - u)$$

$$= \frac{1}{2} (v + u) \quad \text{Since } I = m(v - u)$$

Hence, the work done by the force is equal to the product of impulse and the arithmetic mean of the initial and final velocities.

1. A cricket ball weighing $\frac{5}{16}$ lb is moving with a velocity of 80 ft/s. and is struck by a bat which causes it to travel in the opposite direction with a velocity of 48 ft/s. Then the impulsive force of the bat is [2023]
 1) 10 Poundal/sec 2) 20 Poundal/sec 3) 30 Poundal/sec 4) 40 Poundal/sec.

Ans. Here the initial velocity $= u = 80$ ft/sec.

final $v = v = 48$ ft/sec

$$m = \frac{5}{16} \text{ lb}$$

$$\begin{aligned} \therefore I &= m(v - u) \\ &= \frac{5}{16} (80 + 48) \text{ Poundal/sec} \\ &= \frac{5}{16} \times 128 \text{ Poundal/sec} \\ &= 40 \text{ Poundal/sec.} \end{aligned}$$

2. A cricket ball weighing $\frac{1}{4}$ lb is moving with a velocity of 15 ft/sec and is struck by a bat which causes it to travel in the opposite direction with a velocity of 40 ft/sec. Then the impulsive force of the bat is 20

Ans. Here the initial velocity $= u = 15$ ft/sec

if final $v = v = 40$ ft/sec.

$$m = \frac{1}{4} \text{ lb.}$$

$$I = m(v - u) = \frac{1}{4} (40 - 15) = \frac{1}{4} \times 25 \text{ Poundal/sec.}$$

3. The impulse acting on a body is given by

① impulse = change in K.E ② Impulse = change in Momentum

③ Impulse = Work done by acting force ④ Impulse = None

4. For a rectilinear motion of a particle, if an impulse I changes its velocity from u to v , then the change in K.E is

✓ i) $\frac{1}{2} I(u+v)$ ii) $\frac{1}{2} I(u-v)$ iii) $\frac{1}{2} I(u+v)$ iv) Now —

5. A shell of mass m is fired from a gun of mass M which generates kinetic energy E . Then the initial velocity of the shell is equal to

✓ i) $\sqrt{\frac{2ME}{(M+m)}}$ ii) $\sqrt{\frac{2E}{m+M}}$ iii) $\sqrt{E/M}$ iv) $\sqrt{\frac{2mE}{m+M}}$

1. A shell of mass m is ejected from a gun of mass M by an explosion which generated kinetic energy E .

Prove that the initial velocity of the shell is $\sqrt{\frac{2ME}{(M+m)m}}$.

Ans. let u and v be the velocity of the shell and the velocity of recoil of the gun respectively.

Then by the principle of conservation of linear momentum,

$$Mu + m(-v) = 0$$

$$mu + M(-v) = 0$$

$$u, v = \frac{mu}{M} \quad \text{--- (1)}$$

Total K.E of the shell and the gun is

$$E = \frac{1}{2}mu^2 + \frac{1}{2}Mv^2$$

$$= \frac{1}{2}mu^2 + \frac{1}{2}M \cdot \frac{m^2u^2}{M^2} \quad [\text{by (1)}]$$

$$u, (m+M)mu^2 = 2ME$$

$$u, u = \sqrt{\frac{2ME}{(m+M)m}}$$

Hence the velocity of the shell is $\sqrt{\frac{2ME}{(m+M)m}}$.

2. A shell of mass m is ejected from a gun of mass M by an explosion

2. A gun of mass M fires a shell of mass m horizontally, and the energy of explosion is such as would be sufficient to project the shell vertically to a height h . Show that the velocity of recoil of the gun is $\sqrt{\frac{2mgh}{M(m+M)}}$.

Ans. let E be the energy generated by the explosion. Since the energy is just sufficient to raise a mass m to a height h .

$$E = mgh \quad \text{--- (1)}$$

If u be the velocity of the shell and v that of recoil of the gun.

$$mu = kv$$

$$\therefore u = \frac{kv}{m} \quad \text{--- (2)}$$

Energy of explosion $= E = \frac{1}{2} mu^2 + \frac{1}{2} kv^2$

$$\therefore mgh = \frac{1}{2} mu^2 + \frac{1}{2} kv^2 \quad \text{[by (1)]}$$

$$\therefore 2mgh = m \cdot \frac{k^2 v^2}{m^2} + kv^2 \quad \text{[by (2)]}$$

$$\therefore v^2 = \frac{2mgh}{k^2(m+m)}$$

$$\therefore v = \left\{ \frac{2mgh}{k^2(m+m)} \right\}^{1/2}$$

3. A body of mass $(m_1 + m_2)$ is split into two parts of masses m_1 & m_2 by an internal explosion which generates kinetic energy E . Show that if after explosion the parts in the same line as before, their relative speed is

$$\left\{ \frac{2E(m_1 + m_2)}{m_1 m_2} \right\}^{1/2}$$

Ans. Let u be the velocity of the mass $(m_1 + m_2)$ before explosion and v_1, v_2 be the velocities of the masses m_1, m_2 respectively after explosion.

Then by the principle of Conservation of linear momentum

$$(m_1 + m_2)u = m_1 v_1 + m_2 v_2 \quad \text{--- (1)}$$

Again by the principle of Conservation of energy

$$E + \frac{1}{2} (m_1 + m_2) u^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\therefore 2E + (m_1 + m_2) u^2 = m_1 v_1^2 + m_2 v_2^2$$

Multiplying both sides by $(m_1 + m_2)$

$$2(m_1 + m_2)E + (m_1 + m_2)^2 u^2 = (m_1 + m_2)(m_1 v_1^2 + m_2 v_2^2)$$

$$\text{or } 2(m_1 + m_2)E + \{m_1 v_1 + m_2 v_2\}^2 = m_1^2 v_1^2 + m_2^2 v_2^2 + m_1 m_2 (v_1^2 + v_2^2) \quad [\text{by } \textcircled{1}]$$

~~$$\text{or } 2(m_1 + m_2)E + \{m_1 v_1 + m_2 v_2\}^2 =$$~~

$$\text{or } 2(m_1 + m_2)E + m_1^2 v_1^2 + m_2^2 v_2^2 + 2m_1 m_2 v_1 v_2 = m_1^2 v_1^2 + m_2^2 v_2^2 + m_1 m_2 (v_1^2 + v_2^2)$$

$$\text{or } 2(m_1 + m_2)E = m_1 m_2 (v_1^2 + v_2^2 - 2v_1 v_2)$$

$$\text{or } (v_1 - v_2)^2 = \frac{2E(m_1 + m_2)}{m_1 m_2}$$

$$\text{or } v_1 - v_2 = \left\{ \frac{2E(m_1 + m_2)}{m_1 m_2} \right\}^{1/2}$$

which is the relative speed of the two masses after explosion.